Payroll Fraud Detection Dashboard

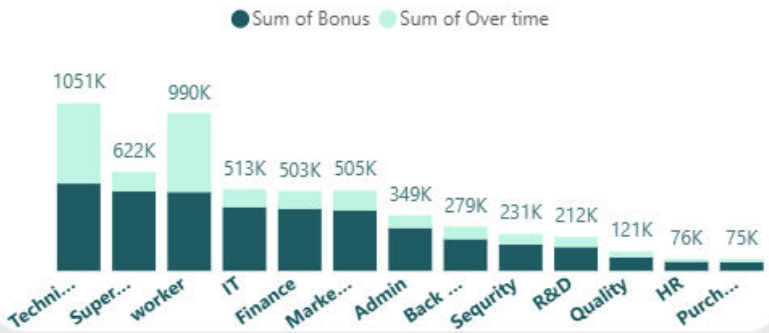
Department

All

Total Employees	Duplicate Employees	Inactive Employees	Suspicious Expenses	Total Expense
1000	608	86	3M	31M

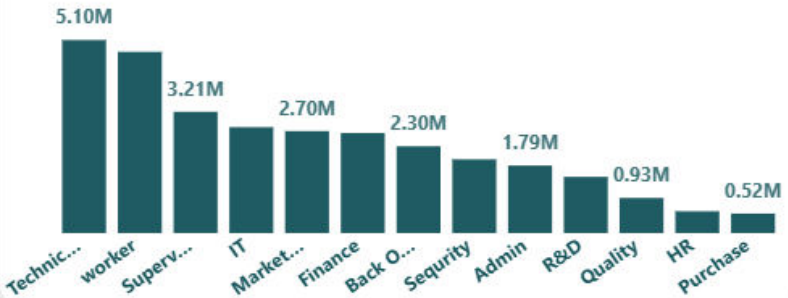
Salary vs Total Expense

Bonus + Overtime Analysis

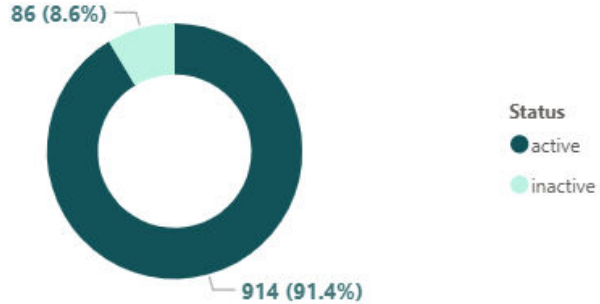


Name	Duplicate	Department
Dina Hassan	9	Finance
Hassan Ibrahim	9	Back Office
Ahmed Farid	8	Back Office
Ali Hassan	8	Admin
Ali Said	8	Back Office
Khalil Farid	8	Back Office
Mohamed Khalil	8	Admin
Noor Fahmy	8	Marketing &
Amr Fahmy	7	Finance
Fatima El-Sayed	7	Marketing &
Hassan Hassan	7	Finance
Ahmed El-Sayed	6	Admin
Ali Khalil	6	Admin
Hassan Farid	6	Finance
Hassan Khalil	6	Admin
Khalil El-Sayed	6	Admin
Khalil Hassan	6	Back Office
Khalil Ibrahim	6	Finance
Lina Hassan	6	HR
Mohamed Farid	6	Admin
Omar Abdelrahman	6	Marketing &
Yang Wei	6	Admin
Total	870	Admin

Total Expense by Department



Employees by Status



Payroll Integrity Audit & Fraud Detection

Financial Analysis & Anomaly Detection using Power BI

The Audit Context

The Challenge

The company suspected salary inflation and payroll irregularities across departments. As part of the financial compliance review, a comprehensive audit was conducted on **1,000 employee records** to identify financial anomalies and detect "Ghost Employees" — individuals receiving compensation despite inactive or terminated status.

Primary Objectives

- Verify payroll accuracy and compliance
- Identify duplicate and ghost employee records
- Quantify suspicious expense patterns
- Provide actionable fraud detection insights

High-Level Insights

The audit reveals significant payroll irregularities requiring immediate management attention.

1000

Total Employees

Complete workforce records audited

$31M

Total Payroll Expense

Annual compensation across all departments

$3M

Suspicious Expenses

Flagged for immediate review

Duplicate Records — The Red Flag

608 Duplicate Names

60.8% of employee records show duplicate entries

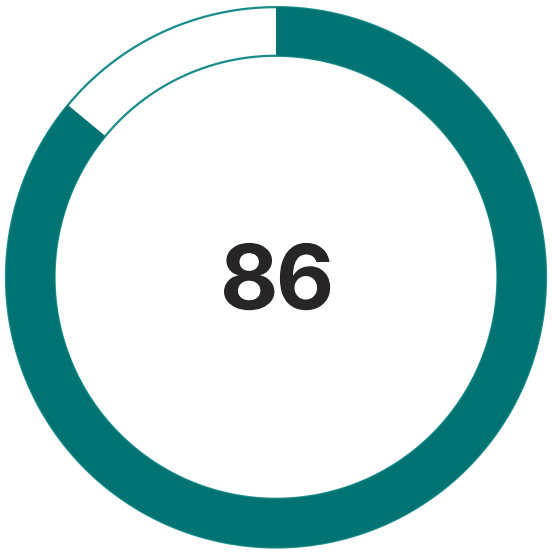
What This Means

The presence of 608 duplicate names suggests potential identity inflation or multiple payroll entries for the same individual. This pattern indicates systematic data entry issues or deliberate attempts to inflate headcount figures.

Next Steps

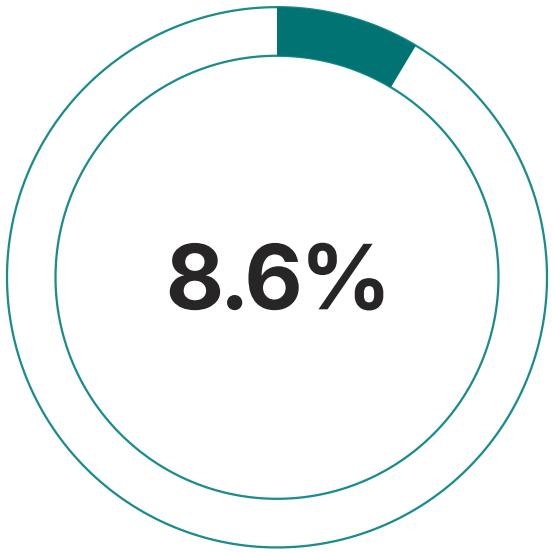
Each duplicate record requires manual verification to confirm legitimate multiple roles versus fraudulent entries. Priority review should focus on duplicates with active payroll status.

Status Inconsistency: Ghost Employees



Inactive Employees

Still receiving compensation



Workforce Impact

Percentage of total employees

Detailed analysis revealed that **86 inactive employees** (8.6% of the workforce) continue to receive active payroll disbursements. These individuals have been terminated or placed on indefinite leave status, yet their records show ongoing salary payments, bonus allocations, and overtime compensation.

Expense Distribution by Department

Payroll expenses are disproportionately concentrated across key operational areas, revealing potential vulnerability zones.

Technology

Highest payroll concentration with elevated bonus and overtime allocations

Operations

Second-highest expense category with inconsistent payment patterns

Supervisory

Disproportionate bonus distribution relative to base salary

Bonus & Overtime Inconsistencies

Bonus Distribution

Total bonus allocations reached **$1.05M** across the organization. Analysis reveals inconsistent award patterns with bonuses distributed to terminated employees and duplicate records.

Overtime Patterns

Overtime hours show abnormal concentration in specific departments with inadequate documentation. Multiple instances of overtime paid to inactive status employees.

Salary vs. Total Expense Analysis

Correlation analysis between base salary and total compensation expense identified clear outliers indicating suspicious financial activity.

Expected Correlation

Total expense should align proportionally with base salary levels across comparable positions

Identified Outliers

Significant deviations where total expense far exceeds base salary, indicating unapproved bonuses or overtime

Direct Evidence

Outlier records provide specific documentation of fraudulent compensation patterns requiring investigation

Data Analysis Methodology

01

Data Collection

Gathered complete employee records including names, employee IDs, department assignments, salary figures, bonus payments, overtime hours, and employment status flags

02

Record Matching

Implemented fuzzy matching algorithms to identify duplicate names and cross-referenced employee IDs to detect multiple entries per individual

03

Status Verification

Filtered active versus inactive employee status codes and validated against HR termination records to identify ghost employees

04

Expense Analysis

Calculated total compensation (salary + bonuses + overtime) and performed correlation analysis to identify outlier expense patterns

05

Reporting

Compiled flagged records with supporting evidence including duplicate entries, inactive payments, and expense outliers for management review

Conclusion & Recommendations

Key Findings

The data confirms a **high risk of payroll fraud** requiring immediate management intervention.

608 Duplicate Records

Require manual verification and system cleanup

86 Inactive Employees

Immediate payroll suspension and investigation

$3M Suspicious Expenses

Detailed audit trail and financial recovery

Next Steps

- Launch formal investigation into flagged records
- Implement automated payroll validation system
- Conduct quarterly compliance audits
- Review and strengthen internal controls